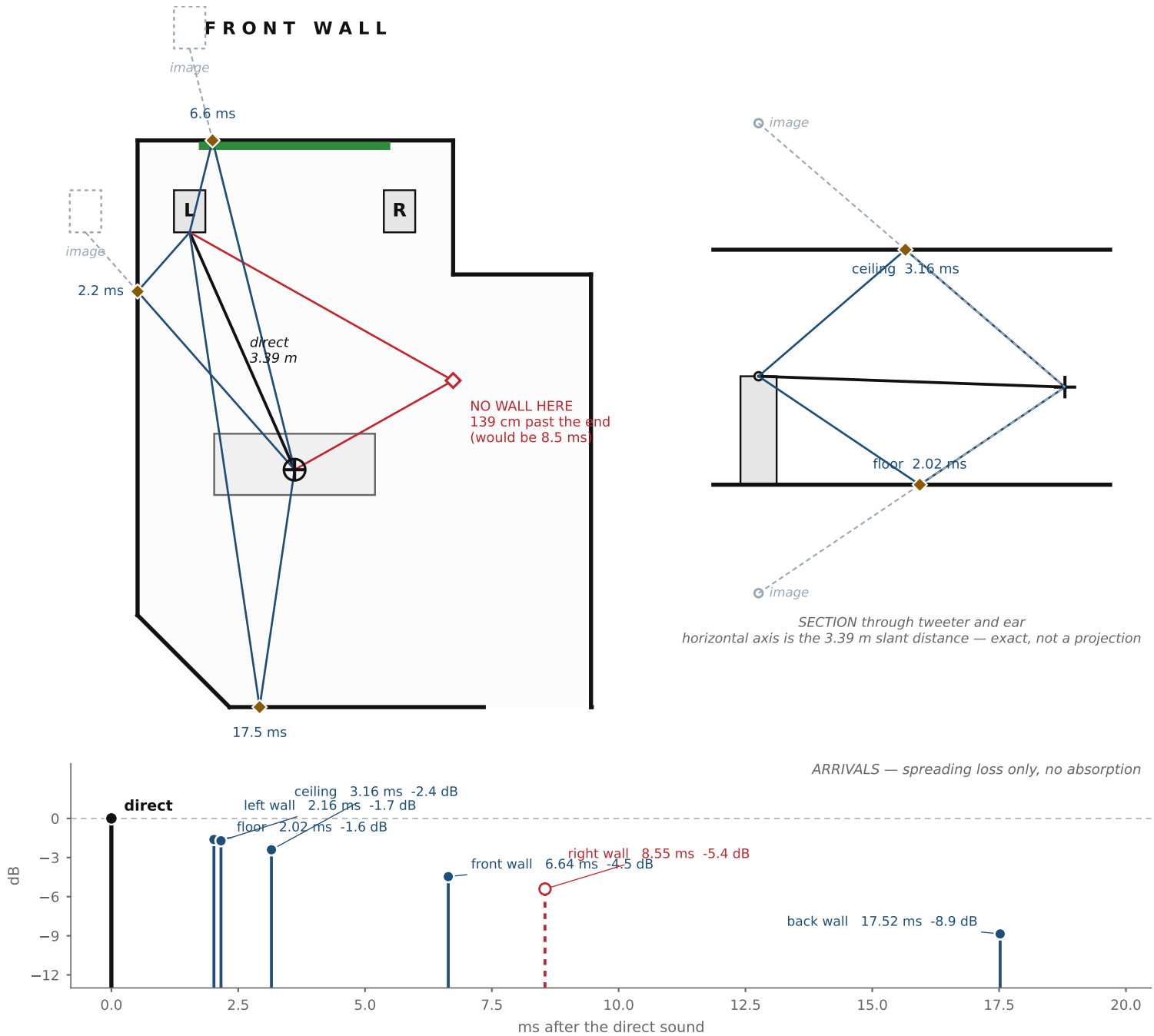


Mirror the source in the surface — the reflected path is then the straight line from image to ear.

— real path — mirror image + its straight line — reflection point — no surface there



surface	image at (x, y, z)	path	excess	delay	level	lands at
direct		3.391 m				
left side wall	(-0.68, +1.20, +1.20)	4.132 m	+0.741 m	2.16 ms	-1.72 dB	y = 1.97 m
right side wall	(+7.56, +1.20, +1.20)	6.323 m	+2.932 m	8.55 ms	-5.41 dB	y = 3.14 m — 139 cm past the half-wall
front wall	(+0.68, -1.20, +1.20)	5.669 m	+2.278 m	6.64 ms	-4.46 dB	x = 0.98 m (1.07 m off centre)
back wall	(+0.68, +13.60, +1.20)	9.401 m	+6.010 m	17.52 ms	-8.86 dB	x = 1.59 m (0.46 m off centre)
floor	(+0.68, +1.20, -1.20)	4.085 m	+0.693 m	2.02 ms	-1.62 dB	2.83 m from the front wall
ceiling	(+0.68, +1.20, +4.00)	4.474 m	+1.082 m	3.16 ms	-2.41 dB	2.69 m from the front wall

Reading it

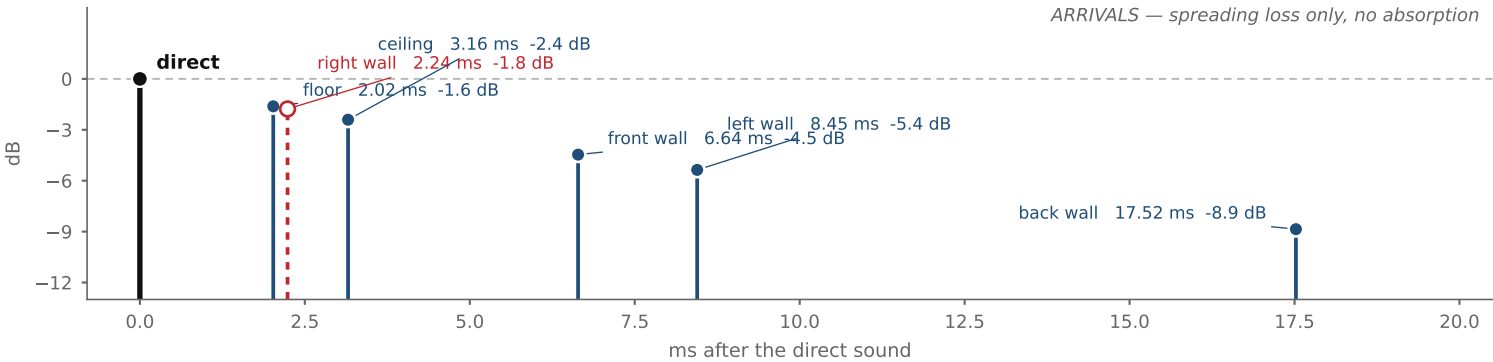
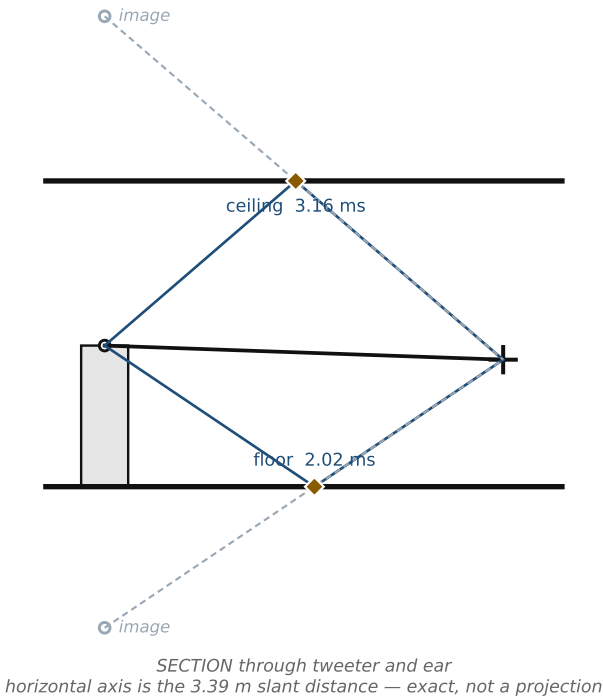
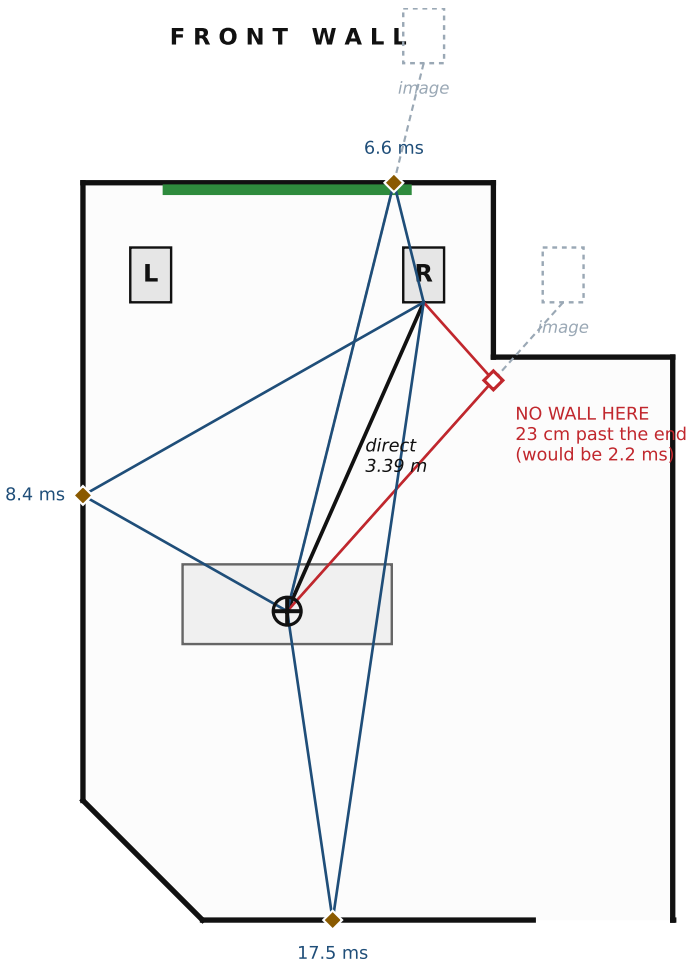
- level is spreading loss only — subtract whatever the surface absorbs, and the speaker's own off-axis roll-off on top of that
- the RIGHT side wall stops 1.75 m from the front wall, so both of its mirror points have no surface to happen on
- floor: the drivers sit at different heights, so it is not one arrival — woofers ~1.1 ms, tweeter 2.0 ms
- ceiling assumed flat at 2.6 m; the real one slants 2.4 → 3.0 m and is beamed, so above ~800 Hz it scatters rather than mirrors

RIGHT speaker — first reflections

Mirror the source in the surface — the reflected path is then the straight line from image to ear.

— real path — mirror image + its straight line — reflection point — no surface there

FRONT WALL



surface	image at (x, y, z)	path	excess	delay	level	lands at
direct		3.391 m				
left side wall	(-3.42, +1.20, +1.20)	6.289 m	+2.897 m	8.45 ms	-5.36 dB	y = 3.14 m
right side wall	(+4.82, +1.20, +1.20)	4.159 m	+0.768 m	2.24 ms	-1.77 dB	y = 1.98 m — 23 cm past the half-wall
front wall	(+3.42, -1.20, +1.20)	5.669 m	+2.278 m	6.64 ms	-4.46 dB	x = 3.12 m (1.07 m off centre)
back wall	(+3.42, +13.60, +1.20)	9.401 m	+6.010 m	17.52 ms	-8.86 dB	x = 2.51 m (0.46 m off centre)
floor	(+3.42, +1.20, -1.20)	4.085 m	+0.693 m	2.02 ms	-1.62 dB	2.83 m from the front wall
ceiling	(+3.42, +1.20, +4.00)	4.474 m	+1.082 m	3.16 ms	-2.41 dB	2.69 m from the front wall

Reading it

- level is spreading loss only — subtract whatever the surface absorbs, and the speaker's own off-axis roll-off on top of that
- the RIGHT side wall stops 1.75 m from the front wall, so both of its mirror points have no surface to happen on
- floor: the drivers sit at different heights, so it is not one arrival — woofers ~1.1 ms, tweeter 2.0 ms
- ceiling assumed flat at 2.6 m; the real one slants 2.4 → 3.0 m and is beamed, so above ~800 Hz it scatters rather than mirrors